
Beyond the Model: How LLM Performance Varies with Harness Design on the 2026 International Mathematical Olympiad

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Abstract

We evaluate five language models on the six fresh problems of the 2026 International Mathematical Olympiad across a web interface, a provider coding agent, and a multi-agent control harness AutoFyn. Internet access was disabled in every setup. Two systems, GPT-5.6 Sol and Claude Fable 5, produced a complete proof of every problem they returned gradable output on, with GPT-5.6 Sol losing points only to locally repairable defects on Problem 2. The other three, Claude Opus 4.8, Claude Sonnet 5, and GLM 5.2, did not, and their total scores move by more than a medal band with the harness they run through. Analyzing the failures, we observe that no sub-frontier run solves Problem 3, a hard combinatorics problem, in any of the eighteen agent runs. Additionally, most models tend to make small but fixable mistakes on Problem 2, which every run approached through long exact computation. We also observe that model and human difficulty rankings diverge, with the models finding Problem 6 easier than Problem 2 and Problem 5 easier than Problem 4, in both cases the reverse of the human order. Since the harnesses differ in tools and prompts, the comparison across harnesses is observational and we make no claim about which harness is best. Every proof was graded under a strict completion standard by a frontier model other than its author and then reviewed by a panel of three past IMO medalists, and every graded write-up and audit report, together with the full process trails of the AutoFyn runs, is archived publicly at <https://github.com/SignalPilot-Labs/IMO-2026>.

1 Introduction

The International Mathematical Olympiad (IMO) is one of the hardest tests of human mathematical reasoning. In July 2026, six hundred and sixty-six of the strongest high school students in the world sat six problems over nine hours, and only seven of them reached a perfect 42/42 ([International Mathematical Olympiad, 2026](#)). Whether AI can reach the same level of performance remains an open question.

In 2024, AI reached the silver threshold ([Hubert et al., 2025](#)), and by 2025 general-purpose models were reported at gold level in natural language ([OpenAI, 2025](#); [Google DeepMind, 2025](#); [Huang & Yang, 2025](#)). Gold is now the year-old bar, and the frontier models have advanced further still. It is therefore worth asking whether AI can produce a flawless paper on the fresh problems of IMO 2026, and how much of any such result belongs to the model itself rather than to the harness it runs through.

We study both questions on the six problems of the 2026 IMO, released after every model’s training cutoff and attempted with web search disabled. We run five models through three harnesses. The first two are the provider’s web interface and the provider’s own coding agent, which are how most users reach these models

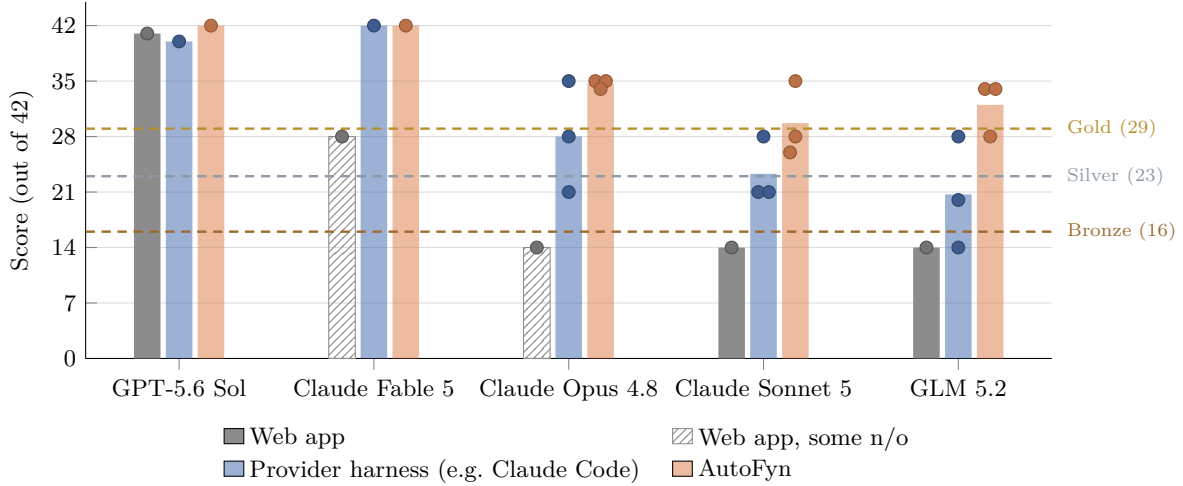


Figure 1: Audited score out of 42 for each model under each harness. Each bar is the mean over that cell’s runs and each circle is one individual run, and the dashed lines mark the 2026 medal cutoffs. Hatched bars are web runs holding at least one problem for which the interface returned no output after the initial attempt and five retries, and each such problem contributes zero (Section 5.2).

in practice. Because these harnesses differ among providers in their system prompts, tools, and available skills, their scores are not mutually comparable. To perform a controlled comparison, we add a third harness, AutoFyn (Hasan et al., 2026), an open-source multi-agent system that we hold fixed, with the same prompt, tools, and retrieval corpus, across all five models. This allows a controlled comparison of the model backends, while the web and provider harnesses together show how each model behaves in the settings a user actually encounters.

We grade every proof under a single strict standard that awards full marks only for a complete argument and zero for any unproved load-bearing step, or only a correct final answer without full proof. To keep the grading honest, each run is audited by a frontier model other than its author, and the scores are revised by a panel of three past IMO medalists, including one gold medalist. The scores we report are this panel’s assessment rather than official IMO grades, and we describe how the standard is applied, and what it costs a near-complete proof, in Section 3.3.

Under this standard, GPT-5.6 Sol and Claude Fable 5 solve all six problems through every harness that returns gradable output, though two of GPT-5.6 Sol’s solutions carry minor repairable defects. The three sub-frontier models, in contrast, miss one or more problems completely, and each of their scores varies widely with the harness. GLM 5.2, for example, scores 14/42 through the web interface and averages 20.7/42 through its provider’s agent and 32.0/42 through AutoFyn, moving more than a medal band with no change to the underlying model. Since prompts, tools, and orchestration vary all at once across harnesses, however, this comparison is observational and we make no claim about which harness is best (Section 3.2).

Under the fixed AutoFyn harness, where the comparison is controlled, all three sub-frontier models land within five points of one another and below the two frontier models. Both frontier models perform strongly in every harness that returned output, and we did not replicate those cells. Figure 1 shows the pattern for every model.

We observe that no sub-frontier run solves Problem 3 in any harness. The problem asks for a minimax value, and a complete proof needs both bounding directions. No sub-frontier run proves both, while the frontier models prove them in every gradable run. We also notice that Problem 6 has a close analogue in a past IMO Shortlist problem, and that every AutoFyn trail surfaces and uses it. The web and provider harnesses carry no retrieval corpus, and both frontier models complete Problem 6 there without reference to a past problem. We read these as observations in the recorded traces but not as isolated causes. We leave separation of the factors to future work.

We also observe that the relative difficulty of the six problems differs between the models and the human contestants (Figure 2). Problem 6 was the second hardest problem for the contestants and was fully solved by only 23 of the 666 participants, yet the sub-frontier models complete it in seven of nine runs under our control harness AutoFyn. Similarly, all nine sub-frontier AutoFyn runs solve Problem 5, a higher rate than they reach on Problem 4, which was far easier for the human contestants. The two rankings do agree at the hard end, where Problem 3 was the least solved by the contestants and goes unsolved in every sub-frontier run.

The rest of the paper is structured as follows. Section 2 gives a brief overview of the related work and Section 3 describes the evaluation matrix, the grading standard, and the audit. Section 4 presents the audited results, and Section 5 presents the analysis of where the systems fail, and how problem difficulty ranks differently for human contestants and for the models.

2 Related Work

It is known in various domains that a fixed model’s score depends on the scaffold around it. For instance, in software engineering the agent-computer interface governs task success (Jimenez et al., 2024; Yang et al., 2024; Shinn et al., 2023), and in mathematics a verification-and-refinement pipeline lifts fixed models from roughly 20–40% to gold on IMO 2025 (Huang & Yang, 2025). These results report what one scaffold achieves, so a score still carries the model and the scaffold together. Concurrent work makes the model-harness pair explicit, benchmarking harness configurations across models in agent workflows (Yao et al., 2026) and transferring one optimized retrieval harness across five held-out models on olympiad-level problems (Lee, 2026). We differ in studying full proofs on a fresh olympiad, in the harnesses users actually reach these models through, and under independent medalist review. The multi-agent harness in our matrix, AutoFyn (Hasan et al., 2026), is an open system that assigns separate subagents to explore approaches, retrieve analogous past problems, prove, and review before a proof is finalized. We use it as the one harness held fixed across every model (Section 3.2).

As frontier models get smarter, the standard math benchmarks are getting saturated and contamination-prone (Hendrycks et al., 2021; Cobbe et al., 2021; Mirzadeh et al., 2024). This makes measuring model capabilities harder, which has driven evaluation onto fresh competitions (Glazer et al., 2024; Balunović et al., 2025). Grading full proofs by model is itself unreliable, since models are biased as judges (Zheng et al., 2023) and miss hard-to-detect proof errors even when the surface reads well (Dekoning et al., 2025).

Against this, our unit of study is the model-and-harness system rather than the model, our contest is fresh and post-training-cutoff-date, and our grades come from an independent cross-model and human-medalist audit rather than a vendor or a self-assessment.

3 Setup

3.1 The 2026 International Mathematical Olympiad (IMO)

The 2026 IMO was held in Shanghai, China, in July 2026. The contestants were given six problems over two days, three per 4.5-hour session, and each problem was marked out of 7 for a maximum of 42. Within a day, the problems were ordered by intended difficulty, so Problems 1 and 4 were meant to be the easiest and Problems 3 and 6 the hardest. In total, 666 contestants competed, and the medal cutoffs were 16 for bronze, 23 for silver, and 29 for gold (International Mathematical Olympiad, 2026). Table 1 gives each problem’s subject area and how many contestants earned a perfect score on it.

3.2 The Evaluation Matrix

We evaluate five models, Claude Opus 4.8, GPT-5.6 Sol, Claude Fable 5, Claude Sonnet 5, and the open-weight GLM 5.2, across three harnesses in a 15-cell evaluation matrix. The first harness is the web application. The second is the provider harness, the provider’s own local agent, Claude Code (v2.1.209) for the Claude models, the Codex CLI (v0.142.5) for GPT-5.6 Sol, and Zcode (v3.4.2) for GLM 5.2. The third is AutoFyn,

Table 1: Perfect score distribution of the six problems of IMO 2026 (International Mathematical Olympiad, 2026).

	Domain	Perfect scores	Percentage
P1	Number theory	443	66.5%
P2	Geometry	100	15.0%
P3	Combinatorics	13	2.0%
P4	Combinatorics	339	50.9%
P5	Algebra	151	22.7%
P6	Number theory	23	3.5%

an open multi-agent harness that explores several approaches per problem, retrieves analogous past olympiad problems, proves, and reviews before finalizing a proof (Hasan et al., 2026). Web access is disabled in every harness, through each harness’s own mechanism (Section A.1). We include AutoFyn as a control harness we hold fixed across all five models, enabling us to compare the model backends under one setting. We make no claim about which harness is best, since ranking harnesses is not the aim of this study, and we note that other multi-agent harnesses exist. For the three sub-frontier models we run both the provider-agent and the AutoFyn cell three times and report each as the mean over its runs, with the individual runs given in Table 3. Since the frontier models solve each of the IMO problems in every gradable output across three harnesses, the experiments were not repeated.

3.3 Grading and Independent Audit

We score all cells under one strict completion-based standard. We give a complete proof 7, and a proof missing a load-bearing component, such as an entire upper or lower bound, 0. We only give partial credit if a proof is complete in substance, with no load-bearing step missing, but has a typo or an unproved minor step whose repair is obvious. There we deduct one or two points after panelist discussion, so the scale is effectively 7 or 0 with 5 and 6 reserved for locally repairable defects. A problem that returned no output scores zero in the run’s total, as it would in the contest. We mark it *n/o* in the per-problem table to distinguish a technical non-submission from a submitted proof that failed.

We adopt this near-binary standard rather than graduated partial credit because partial credit requires the grader to assign a fair value to an incomplete argument. In practice, this is difficult to determine consistently and increases the burden of reaching agreement between AI and human graders. Restricting the decision to the completeness and soundness of the proof makes the grade reproducible and admits a unanimous panel verdict.

To avoid self-grading, each archived run is audited by a distinct frontier model under a fixed reviewer prompt, with Claude and GLM runs audited by GPT-5.6, and GPT-5.6 runs audited by Claude Fable 5. The model audits are then reviewed by a panel of three past IMO medalists with a gold medalist as leader. The scores reported here are therefore the model audit together with this medalist review, and are not official IMO jury grades.

We find that the panelist review was indeed necessary. Where the panel disagreed with a model audit, it did so by finding a mistake in the proof that the auditor had missed, and deducted points accordingly. The scores in Table 2 are therefore the outcome of both stages, and we archive each run and its audit report so that the audit can be verified independently.

3.4 On Comparing Harnesses

Each cell runs a harness as shipped, at its default configuration, the operating point a user gets out of the box. We impose only two uniform limits on top of each harness’s own defaults: a budget of 20 hours per

Table 2: Per-problem scores out of 7, graded as in [Section 3.3](#). Sub-frontier agent cells report the mean over three runs, so those entries may be fractional. A problem marked “n/o” returned no output and scores zero in the total.

Model	Harness	P1	P2	P3	P4	P5	P6	Total
GPT-5.6 Sol	Web (chatgpt.com)	7	6	7	7	7	7	41/42
GPT-5.6 Sol	Codex	7	5	7	7	7	7	40/42
GPT-5.6 Sol	AutoFyn	7	7	7	7	7	7	42/42
Claude Fable 5	Web (claude.ai)	7	7	n/o	7	7	n/o	28/42
Claude Fable 5	Claude Code	7	7	7	7	7	7	42/42
Claude Fable 5	AutoFyn	7	7	7	7	7	7	42/42
Claude Opus 4.8	Web (claude.ai)	n/o	n/o	n/o	7	7	n/o	14/42
Claude Opus 4.8	Claude Code	7.0	4.7	0.0	7.0	7.0	2.3	28.0/42
Claude Opus 4.8	AutoFyn	7.0	6.7	0.0	7.0	7.0	7.0	34.7/42
Claude Sonnet 5	Web (claude.ai)	7	0	0	0	7	0	14/42
Claude Sonnet 5	Claude Code	7.0	2.3	0.0	4.7	7.0	2.3	23.3/42
Claude Sonnet 5	AutoFyn	7.0	4.3	0.0	6.7	7.0	4.7	29.7/42
GLM 5.2	Web (chat.z.ai)	7	0	0	7	0	0	14/42
GLM 5.2	Zcode	7.0	4.7	0.0	4.3	4.7	0.0	20.7/42
GLM 5.2	AutoFyn	7.0	6.3	0.0	7.0	7.0	4.7	32.0/42

problem, and, when a system returns no output on its first attempt, up to five retries on the same problem before we record the problem as *n/o* and score it zero. We check the budget between rounds rather than interrupting a round in progress, so a single Problem 6 run overran it and finished at just over 20 hours of active work. The retry rule matters mainly for the Claude web interfaces, where a single attempt can terminate early on a message-length limit rather than on the mathematics ([Section 5.2](#)). A run ends when the system submits its final write-up, exhausts its retries, or reaches the budget. Technical failures, such as the web application’s message-length cap, are scored as zero, and recorded as such ([Section 5.2](#)).

The two directions of the matrix carry different evidential weight. Comparing models within a single harness holds the harness, its version and defaults, the problems, and the grading fixed, so these comparisons are controlled; the five AutoFyn rows of [Table 2](#) differ only in the model. Comparing harnesses for a fixed model spans cells that differ on tools, orchestration, and effective budget at once, so these comparisons are descriptive; we report how outcomes vary with the harness without attributing the difference to any single ingredient. Episode length is the largest of these differences, and [Table 4](#) shows it varies by up to thirteenfold across the harnesses a single model runs through.

4 Results

[Table 2](#) gives the audited scores on the six fresh problems of the 2026 IMO, under the completion-based standard of [Section 3.3](#), with each three-run cell reported as its mean. [Table 3](#) then breaks the six repeated cells into their individual runs, and [Table 4](#) records episode lengths for the same cells. We note that in Claude’s web harness some problems returned no output at all, and we score those zero ([Section 5.2](#)).

Table 3: Perfect score by problem breakdown for the six sub-frontier agent cells, each run three times independently.

Model	Harness	P1	P2	P3	P4	P5	P6	Runs	Average
Claude Opus 4.8	Claude Code	3/3	2/3	0/3	3/3	3/3	1/3	21, 35, 28	28.0/42
Claude Opus 4.8	AutoFyn	3/3	2/3	0/3	3/3	3/3	3/3	35, 35, 34	34.7/42
Claude Sonnet 5	Claude Code	3/3	1/3	0/3	2/3	3/3	1/3	21, 28, 21	23.3/42
Claude Sonnet 5	AutoFyn	3/3	1/3	0/3	2/3	3/3	2/3	35, 26, 28	29.7/42
GLM 5.2	Zcode	3/3	2/3	0/3	1/3	2/3	0/3	20, 28, 14	20.7/42
GLM 5.2	AutoFyn	3/3	1/3	0/3	3/3	3/3	2/3	34, 34, 28	32.0/42

Table 4: Episode length in minutes, averaged over the same runs as Table 2. The AutoFyn figures run longer because of its multi-agent and multi-round approach. Runtimes are not comparable across harnesses (Section 3.4). We exclude idle time spent awaiting a rate-limit reset.

Model	Harness	P1	P2	P3	P4	P5	P6	Total
GPT-5.6 Sol	Web (chatgpt.com)	3	23	13	5	3	15	62
GPT-5.6 Sol	Codex	3	13	14	7	4	10	51
GPT-5.6 Sol	AutoFyn	18	55	56	23	24	60	236
Claude Fable 5	Web (claude.ai)	3	65	157	11	8	47	291
Claude Fable 5	Claude Code	5	19	37	20	4	38	123
Claude Fable 5	AutoFyn	26	51	92	49	27	155	400
Claude Opus 4.8	Web (claude.ai)	51	32	79	48	10	80	300
Claude Opus 4.8	Claude Code	4	118	78	35	32	119	386
Claude Opus 4.8	AutoFyn	21	78	543	60	29	142	873
Claude Sonnet 5	Web (claude.ai)	5	17	24	64	43	17	170
Claude Sonnet 5	Claude Code	3	36	69	54	102	64	328
Claude Sonnet 5	AutoFyn	39	327	906	59	27	842	2200
GLM 5.2	Web (chat.z.ai)	9	11	24	19	13	34	110
GLM 5.2	Zcode	9	33	240	33	56	128	499
GLM 5.2	AutoFyn	25	200	433	56	51	219	984

In Table 2, we observe that GPT-5.6 Sol and Claude Fable 5 complete every problem on which they returned gradable output. Claude Fable 5 earns full marks throughout, while GPT-5.6 Sol loses points only on Problem 2, in each case to a locally repairable defect rather than a missing step. In contrast, the other three models do not complete a full contest in any harness, and their totals vary by more than a medal band across the harnesses. Claude Sonnet 5, for instance, scores 14/42 on the web and 29.7/42 under AutoFyn without any change to the model. We report that span as an observation about deployments rather than directly ranking harnesses (Section 3.4).

In our control harness AutoFyn, the three sub-frontier models fall within five points of one another: Claude Sonnet 5 at 29.7/42, GLM 5.2 at 32.0/42, and Claude Opus 4.8 at 34.7/42, and all three sit below the two frontier models. The individual runs of these cells range from 26 to 35 (Table 3), so the spacing between the means is well inside the spread within them.

We notice that no sub-frontier run solves Problem 3 in any harness, while both frontier models complete it wherever they returned output. We also find that Problem 2 fails differently and costs every model besides

Fable points in at least one run. Every model attempts it with a computational approach and needs to perform long exact calculation. The proofs then fail by small algebraic mistakes rather than outright. Six of the eight partial scores awarded anywhere in the matrix fall on Problem 2, including the only points GPT-5.6 Sol loses in any harness. [Section 5](#) takes up both patterns.

5 Analysis

We ran each of the three sub-frontier models three times in its provider agent and three times in AutoFyn.

5.1 How Proofs Fail

Problem 3 asks for a minimax value, and a complete proof needs to close both bounding directions. While both frontier models complete it in every run where they returned gradable output, none of those eighteen sub-frontier runs close it in both directions.

Two runs close exactly one direction, namely Claude Sonnet 5 proving the lower bound in full in its first AutoFyn run and Claude Opus 4.8 the upper bound in full in its second, and the remaining sixteen close neither. We do not read these as runs that ran short of time, since the longest single run worked the problem for over nineteen hours.

The remaining problems fail in distinguishable ways. We find that Problem 2 admits the most small and local mistakes. Every model reduces it to a long exact calculation, and we see six of the eight partial scores awarded anywhere in the evaluation matrix fall on it. We also note it is the only problem on which GPT-5.6 Sol loses points at all.

Furthermore, we notice that Problem 6 fails at one identifiable step. It has a close analogue in a past IMO Shortlist problem, 2013 N5, and we see every AutoFyn trail surface and use it. Seven of the nine sub-frontier runs close the proof and two do not, and we find both failures stalling at the passage from local structure to a finite-state process.

Only AutoFyn carried its internal retrieval corpus. In the web and provider harnesses, we see that both frontier models complete the problem without reference to a past problem in every run with gradable output, while only two of the nine sub-frontier provider-harness runs manage the same. We report these differences as observational, since the settings differ in tools, orchestration, and budget at once ([Section 3.4](#)), and we leave the question of what retrieval contributes to a future study.

Lastly, we find that Problem 4 requires an exhaustive case-by-case analysis rather than a single idea, and four of the eighteen sub-frontier runs fail to carry it through. Two of these runs reach the correct characterization and need only a local repair, scoring a 6, while the other two miss the characterization itself, one omitting the converse direction entirely and one proposing an invalid strategy.

5.2 Technical Failures

We observe that a harness can also hold a model back. In particular, on the web interface, the Claude Opus 4.8 run hit the message-length limit on four problems and the Claude Fable 5 run on two, returning no output across five retries, so those attempts score zero. Claude Fable 5 scores 42/42 through both agent harnesses and 7/7 on the four web problems it could submit. We therefore find that a model able to complete the whole contest can still be bounded by the interface it is reached through rather than by the problems.

5.3 Human and Model Difficulty

Next, we rank the six problems by their difficulty for human contestants and for the models. We count complete solutions among the number of the nine sub-frontier AutoFyn runs that scored a full 7 and the number of the 666 contestants who scored a perfect 7 at the 2026 IMO. We do not consider mean score because partial-credit conventions differ across problems, so a mean does not compare from one problem to the next while a complete proof is a uniform bar. [Figure 2](#) sets the two side by side, with the problems ordered by human difficulty.

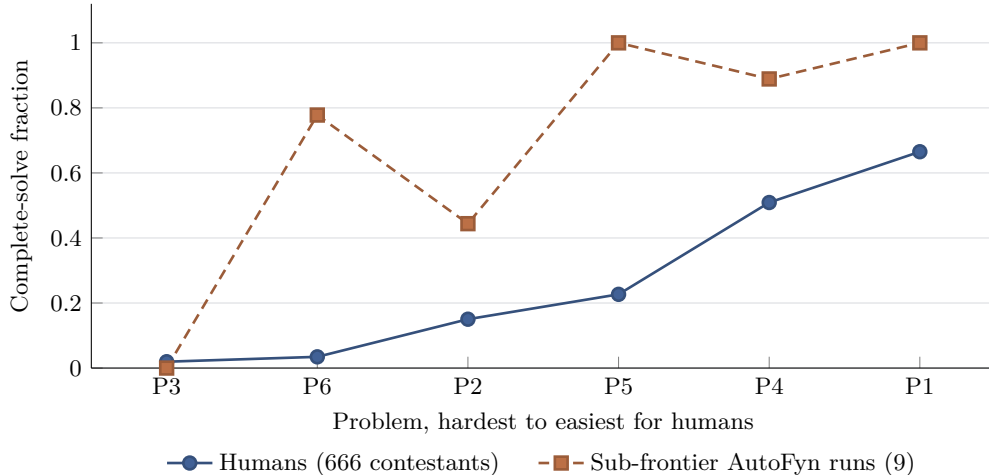


Figure 2: Fraction of each population that solves a problem completely (International Mathematical Olympiad, 2026). The problems are ordered by human difficulty, so the human series rises by construction and any departure from that rise marks a disagreement with the model ordering. The two populations work under different conditions, so the vertical scale is not a hardness equivalence between them.

We find the two rankings agreeing at the extremes. The three problems fewest humans solved, Problems 3, 6, and 2, are also the three we see fewest model runs solve, and we find the three problems most humans solved completed in nearly every model run. Within the hard group, however, the rankings differ. On Problem 2 all models attempt a computational solution, and the proofs that fall short miss by a single repairable point. Attempts at Problem 6, in contrast, fail outright. Counting complete solutions, we therefore find Problem 6 easier than Problem 2 for the models, the reverse of the human order.

We also note that Problems 3 and 6 were solved by only 13 and 23 of the 666 contestants. The models split them apart, completing Problem 6 in seven of the nine sub-frontier AutoFyn runs and Problem 3 in none. We analyze how they fail in Section 5.1.

Lastly, in the easy group, we find that the models flip the human difficulty ranking of Problems 4 and 5 and even solve Problem 5 faster than Problem 4 in eight of the nine AutoFyn runs. In future, we aim to isolate the cause of this flip in a larger set of similar problems.

5.4 What the Standard Costs a Near-Complete Proof

Our standard is unforgiving of a near-miss, and we credit neither a correct final answer nor a sound structure built around a hole. We take the two strongest Problem 3 runs as the clearest instance. Claude Sonnet 5’s first AutoFyn run gives the correct value $c(n) = 2^n / (2^{n+1} - 1)$, proves the lower bound in full, and closes the upper-bound induction for every configuration through $m = 4$ before leaving the general case open. Claude Opus 4.8’s second run proves the upper bound in full and leaves two named leaves of the lower bound unclosed. We score both 0, and we accept this cost deliberately. We award full marks only when every load-bearing step is present, and a proof missing one is always graded zero. We adopt the standard because it makes the totals in Table 2 comparable across models and harnesses, leaving us no discretion over what an incomplete argument is worth.

6 Limitations

- **One contest.** We evaluate five models over three harnesses on the fifteen cells of a single olympiad in a single year. The matrix is therefore small, and the trends we report rest on a limited number of problems and cells. Whether they hold on other contests is left to future work.

- **Replication is asymmetric.** We repeat only the six sub-frontier agent cells, three runs each. Every frontier cell and every web cell is a single run and we do not make any claims about how stable those results are.
- **Scores are not official grades.** The reported scores are our panelist assessment and not official IMO jury grades. No model graded its own run, but models remain part of the audit loop.
- **The harness comparison is confounded.** Moving a model from its provider agent to AutoFyn changes tools, orchestration, and effective budget together, so we cannot attribute any score difference to a single ingredient. Hence, we report these differences as observational and draw no causal ranking (Section 3.4).
- **No retrieval ablation in AutoFyn.** The pre-2026 corpus is available in every AutoFyn run, so our design has no no-retrieval arm and cannot measure what retrieval contributes. We note only that on Problem 6 the runs that fail retrieve the same analogue as the runs that succeed.
- **Harness set is not exhaustive.** We report the three harnesses we ran end-to-end under our audit and archival protocol. Others exist, including the verification-and-refinement pipeline of Huang & Yang (2025), and running them under this protocol is a natural extension.
- **Freshness is a protocol claim.** Freshness is a claim about the declared environment and not an absolute guarantee that a problem was novel to a hosted model backend.

7 Conclusion

We graded five models on a fresh olympiad under a strict completion standard, each through three harnesses. Two models complete the contest wherever they returned gradable output. The other three do not, and their observed scores vary substantially with the harness as well as with the model backend. This is why we treat the model-and-harness system rather than the model as the unit of measurement.

In Problem 3, we observe that sub-frontier models lag behind frontier models regardless of harness. This is the one separation in our matrix that the harness does not move. Beyond this problem, we find that model and human difficulty rankings diverge.

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A Appendix

A.1 Reproducibility

We ran every cell between 15 and 28 July 2026, after the contest and after every model’s training cutoff. We disabled web access in each harness through the mechanism that harness provides: the web interfaces expose it as a toggle, the provider agents as a configuration setting, and AutoFyn, which runs inside a Docker container for isolation, by withholding its web access. The tools available to an AutoFyn run are file read, file write, file edit, glob, grep, and Bash, together with AutoFyn’s own parallel approach-ranking manager, which maintains the population of rival approaches a run keeps for its problem.

Table 5: Per-problem scores for all eighteen sub-frontier agent runs, out of 7. Each cell of Table 2 averages the three totals in its block.

Model	Harness	Run	P1	P2	P3	P4	P5	P6	Total
Claude Opus 4.8	Claude Code	1	7	0	0	7	7	0	21
Claude Opus 4.8	Claude Code	2	7	7	0	7	7	7	35
Claude Opus 4.8	Claude Code	3	7	7	0	7	7	0	28
Claude Sonnet 5	Claude Code	1	7	7	0	0	7	0	21
Claude Sonnet 5	Claude Code	2	7	0	0	7	7	7	28
Claude Sonnet 5	Claude Code	3	7	0	0	7	7	0	21
GLM 5.2	Zcode	1	7	7	0	6	0	0	20
GLM 5.2	Zcode	2	7	7	0	7	7	0	28
GLM 5.2	Zcode	3	7	0	0	0	7	0	14
Claude Opus 4.8	AutoFyn	1	7	7	0	7	7	7	35
Claude Opus 4.8	AutoFyn	2	7	7	0	7	7	7	35
Claude Opus 4.8	AutoFyn	3	7	6	0	7	7	7	34
Claude Sonnet 5	AutoFyn	1	7	7	0	7	7	7	35
Claude Sonnet 5	AutoFyn	2	7	6	0	6	7	0	26
Claude Sonnet 5	AutoFyn	3	7	0	0	7	7	7	28
GLM 5.2	AutoFyn	1	7	6	0	7	7	7	34
GLM 5.2	AutoFyn	2	7	6	0	7	7	7	34
GLM 5.2	AutoFyn	3	7	7	0	7	7	0	28

AutoFyn retrieves from a fixed corpus that we archive alongside the runs. It holds 1026 problems from olympiads and national team selection tests between 2006 and 2025, and 2434 load-bearing moves extracted from their solutions. Every entry predates the 2026 contest.

The web and provider-agent cells do not expose a run trail, so for those we archive the graded write-up.

For each AutoFyn run we archive its run log. We take the scores in Table 2 and Table 3 from each run’s audit file, which records a score, a verdict, and the auditor’s reasoning for every problem. We compute the AutoFyn times in Table 4 from the run-log timestamps as the span from the first to the last event, less every idle gap longer than five minutes, so that waiting on a rate-limit reset does not count against a run. We take the web and provider-agent times from the session start and end times instead, since those cells expose no log.

A.2 Per-Run Scores

Table 5 gives the per-problem score of every sub-frontier agent run, from which the means in Table 2 and the solve counts in Table 3 can both be recomputed.

A.3 Solver Prompts

We use two solver prompts. In the web and provider harnesses we issue the following as a single message with the problem statement appended:

Can you use deep think for at least 60 minutes and use any means at your disposal (except web search), to solve the following hard math problem? Try multiple ideas even if you reach

dead ends on some of them. If you find a solution idea, try to formalize into a rigorous solution, and skeptically review it for any gaps or bugs. Do not present partial progress or approaches, only provide a complete rigorous verified solution. <problem>

AutoFyn carries its own instructions in a `CLAUDE.md` file, archived alongside the runs, that the orchestrator reads at the start of a run, so there we issue a shorter prompt:

You are a world class mathematician orchestrating a team to solve olympiad math problems. read CLAUDE.md for instructions and solve the problem. <problem>

We do not expect a rerun to reproduce our scores exactly, since one repeat cell ranges from 26 to 35 under identical conditions. Regrading, however, is exact. We archive every write-up beside the audit that scored it, so the grading can be verified.

A.4 Reviewer Prompt

We audit each run under the following fixed prompt ([Section 3.3](#)):

Can you review these claimed solutions of a hard math contest? Be extremely skeptical and try to find any gaps in the argument or flaws in the proof. Give a final verdict on the score between 0 and 7. Do not give partials if a load-bearing step is missing. Give partials only if the proof is correct but has a locally repairable mistake whose repair is obvious. Where useful, write and run verification code to stress-test load-bearing claims, and do not invent issues that are not there.